Gradient function graphs

1

a 1

b 1

c A linear function has a constant gradient.

2

a The gradient function is positive when $y = x^2$ is increasing, and negative when $y = x^2$ is decreasing.

b Positive

c The gradient of the tangent line increases at a constant rate.

d Negative

e The gradient of the tangent line increases at a constant rate.

f Linear

3

a Positive

b The gradient of the tangent line increases at an increasing rate.

c Positive

d The gradient of the tangent line decreases at an decreasing rate.

e Quadratic

4

a Positive

b The gradient of the tangent line increases at an increasing rate.

c Negative

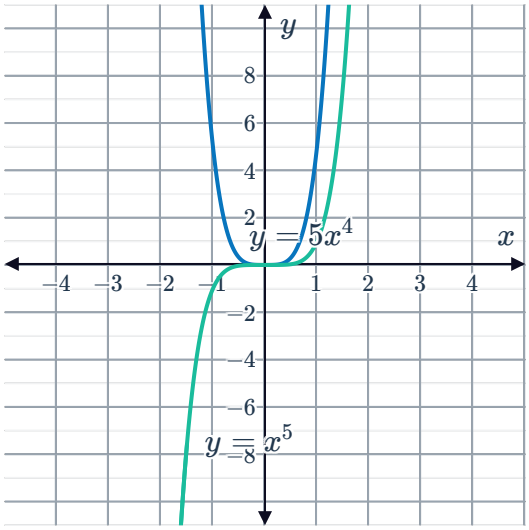
d The gradient of the tangent line increases at a decreasing rate.

e Cubic

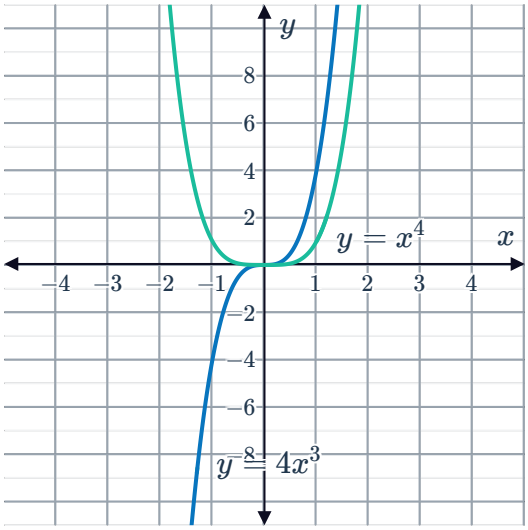
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5

a



b



c Odd

d Even

Algebraic differentiation

6

a $2x$ **b** $8x^7$ **c** nx^{n-1}

7

a

i Degree 1 **ii** Degree 2 **iii** Degree 3

b $n-1$

8

a $\frac{dy}{dx} = 7x^6$ **b** $\frac{dy}{dx} = 9x^8$ **c** $\frac{dy}{dx} = -7x^{-8}$ **d** $\frac{dy}{dx} = -x^{-2}$

e $\frac{dy}{dx} = \frac{6}{5}x^{\frac{1}{5}}$ **f** $\frac{dy}{dx} = \frac{2}{3}x^{-\frac{1}{3}}$ **g** $\frac{dy}{dx} = 0$ **h** $\frac{dy}{dx} = 0.6x^{-0.4}$

9

a $y = x^{\frac{1}{2}}$ **b** $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$

10

a $y = x^{-2}$ **b** $\frac{dy}{dx} = -2x^{-3}$

11

a $y = x^{-\frac{1}{4}}$ **b** $\frac{dy}{dx} = \frac{-1}{4x^{\frac{5}{4}}}$

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12 $\frac{dy}{dt} = \frac{3}{2}t^{\frac{1}{2}}$

13 $\frac{dy}{dx} = -\frac{1}{5x^{\frac{6}{5}}}$

Applications

14 $f'(2) = 32$

15

a 9 **b** $y = 9x - 8$

16

a $-\frac{1}{2}$ **b** $x + 2y - 3 = 0$

17

a

Graph	y-intercept of the tangent line	Gradient of the tangent
$y = x^2$	$(0, -1)$	2
$y = x^3$	$(0, -2)$	3
$y = x^4$	$(0, -3)$	4
$y = x^5$	$(0, -4)$	5

b n

c No, because then the gradient of the tangent at $x = 1$ would be 1.

18

a False **b** True **c** False **d** True

19

a $\frac{4}{3}x^{\frac{1}{3}}$ **b** $\frac{3}{4}x^{-\frac{1}{4}}$

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